



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester I 2025/2026

Subject : Enterprise Systems Design & Modelling (SECSP3744)

Task : SAP BTP INDIVIDUAL

Due : W12 (monitor week by week in Github)

Submission : Upload softcopy short paper (in .pdf format) via e-learning.

File name : G5_LING_YU_QIAN_INDIVIDUAL_SAP

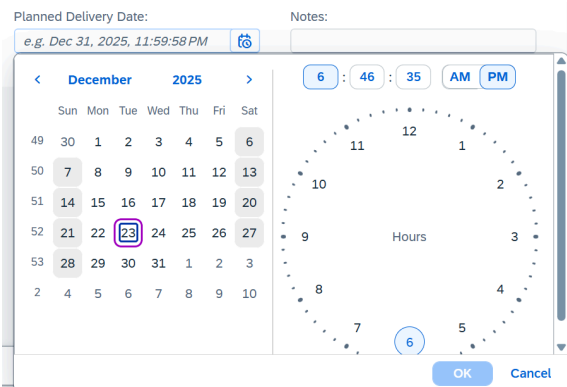
GitHub link: <https://github.com/lingyuqian0301/sapfiori>

Demo link: <https://youtu.be/03ia-qyIvoQ>

1.0 Project Initialization Overview

This project is a Large Item logistics Management Application built with SAP Cloud Application Programming(CAP) Model , SAP Fiori,Odata service. The application was developed and tested in a local environment with SQLite for the back-end as database prior to deployment to SAP BTP.

The project folder that was created is MyHANAApp. This folder saves the database models, service definitions, and config files to run the SAP Cloud Application Programming (CAP).

Figure	Description
 Figure 1.1 Planned Delivery Date Picker	Delivery Date and Time Picker for the logistic product.

Large Item Delivery Details

Delivery ID: A0002

Item ID: P0001

Pickup Location ID: B000

Number of Items: 50

Delivery Status: Pending

[binID]	[location]	[capacity]	[binType]
B0001	Main Warehouse - Zone A	500	Warehouse
B0002	Main Warehouse - Zone B	500	Warehouse
B0003	Distribution Hub North	300	Hub
B0004	Distribution Hub South	300	Hub
B0005	Customer Site - Downtown	50	Customer
B0006	Customer Site - Industrial Park	100	Customer
B0007	Returns Processing Center	200	Warehouse
B0008	Main Warehouse - Zone C	500	Warehouse

Show All Items

Draft updated: [Create](#) [Discard Draft](#)

Figure 1.2 Delivery Details Creation Form

Enter product data, include pickup location ID from Odata that store in the Odata location



Figure 1.3 CRUD Of the system

Can perform create, read, update, and delete (CRUD) in SAP CAP.

View Settings [Reset](#)

Columns Sort Filter Group

Filter By

- comment
- Delivery ID
- Delivery Status
- Dropoff Location ID
- Item ID
- Number of Items
- Pickup Location ID
- transferDate

Figure 1.4 List View Filter Settings

Can filter by the data that created

View Settings

Columns

Sort

Filter

Group

|Sort By| ▼

- comment
- Delivery ID
- Delivery Status
- Dropoff Location ID
- Item ID
- Number of Items
- Pickup Location ID
- transferDate

≡ ≡

Ascending

Figure 1.5 List View Sort Settings

Can sort by data whether ascending or descending that created

< **SAP** Preview – List of StockService.Bins ▼
Standard ▼

Search Q

☐	Bin ID	Location	Capacity	Bin Type	
☐	B0001	Main Warehouse - Zone A	500	Warehouse	>
☐	B0002	Main Warehouse - Zone B	500	Warehouse	>
☐	B0003	Distribution Hub North	300	Hub	>
☐	B0004	Distribution Hub South	300	Hub	>
☐	B0005	Customer Site - Downtown	50	Customer	>
☐	B0006	Customer Site - Industrial Park	100	Customer	>
☐	B0007	Returns Processing Center	200	Warehouse	>
☐	B0008	Main Warehouse - Zone C	500	Warehouse	>

Figure 1.6 Warehouse Bin Locations

Data location that store the Pick Up location and also the drop-off location

 Preview – List of StockService.Products ▾

Search In: "Apps"

Standard ▾

Search

☐ Product ID	Name	Category	Unit of Measure	
☐ P0001	Large Sofa Set	Furniture	pcs	>
☐ P0002	Refrigerator 500L	Appliances	pcs	>
☐ P0003	Dining Table Set	Furniture	set	>
☐ P0004	Washing Machine	Appliances	pcs	>
☐ P0005	King Size Bed Frame	Furniture	pcs	>
☐ P0006	55" Smart TV	Electronics	pcs	>
☐ P0007	Office Desk	Furniture	pcs	>
☐ P0008	Dishwasher	Appliances	pcs	>
☐ P0009	Wardrobe	Furniture	pcs	>
☐ P0010	Air Conditioner	Appliances	pcs	>

Figure 1.7 Product Master List

Data location that store the product

 Preview – List of StockService.Status ▾

Standard ▾

Search

☐ Status ID	Description	
☐ Cancelled	Cancelled	>
☐ Delivered	Delivered	>
☐ InTransit	In Transit	>
☐ Pending	Pending Approval	>
☐ Returned	Returned	>
☐ Scheduled	Scheduled for Transfer	>

Figure 1.8 Delivery Status Definitions

Data location that store the status

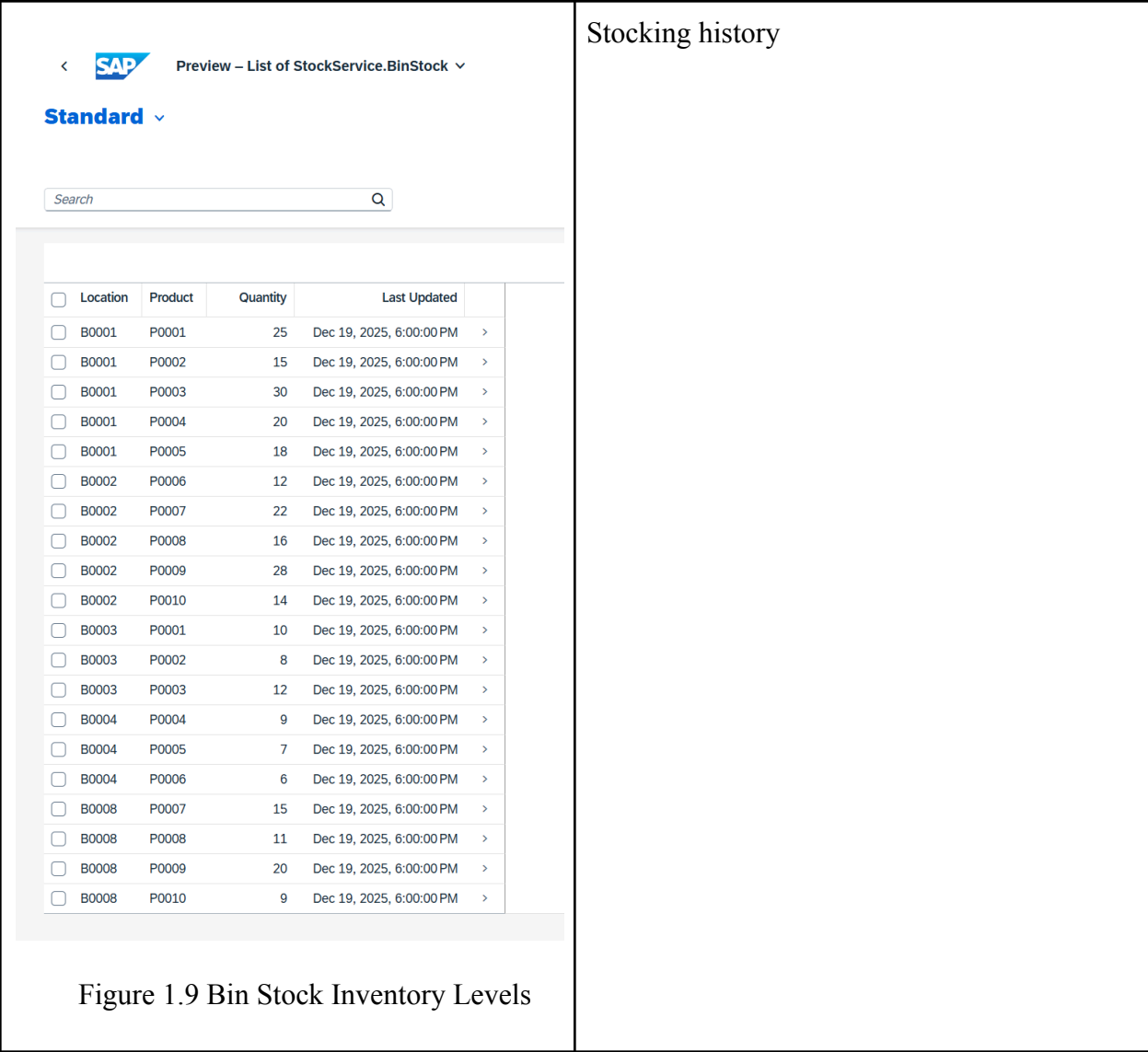


Table 1.1.1 Step of Project Initialization

2.0 Application Initialization

2.1 Environment Setup

The configuration was verified to ensure the database connection was correct.

In the package.json file, the database configuration is under cds.requires. db was set to use a local SQLite file: Database URL: db.sqlite
(This step is to avoid database deployment errors.)

2.2 Running the Application

The following steps were executed in sequence:

1) Install dependencies	All required Node.js libraries(node module) were installed using: npm install
2)Deploy database models	The CDS models were deployed to SQLite, creating the database tables: cds deploy --to sqlite
3) Start the CAP server	The application server was launched using: cds watch

Table 2.2.1 Step of running the sap application (CAP)

2.3 Testing the Application

After the server started successfully:

1)Go to http://localhost:4004

2)Open logistics service
3)launch Fiori Preview launched to display the interface
4) start to create a logistic product in the preview
5) it will successfully created and show in terminal as well as fiori interface
6)backend service run successfully

Table 2.3.1 Step of testing the sap application (CAP)

3. Assessment Documentation

3.1 What tutorial you follow?

I follow a tutorial to build a large-item logistics management application by SAP Cloud Application Programming (CAP) Model, SAP Fiori, and OData. It is due to the reason that all of them are one of the beginner-level services in SAP BTP. I am also utilise CDS models and OData services and testing using SQLite.

3.2 What you explored

I explored how the SAP CAP Model and SAP Fiori work within SAP BTP by :

1)creating backend services	logistic transfer management
2)defining database entities	logistic records and administrative data
3)connecting the application to an SQLite database	development and testing.
4) Using Sap Fiori element for ui	Sap logistic interface and preview

Table 3.2.1 Exploring sap btp

3.3 What steps you tried

Data Modeling	Defined database entities in the db/ folder, including entities related to stock transfers and administrative data.
Service Creation	Exposed the database entities through service definitions in the srv/ folder so they can be accessed by the frontend.
Database Deployment	Used cds deploy --to sqlite to generate database tables from CDS models.
Sap Fiori	Using Sap Fiori element for ui design
Odata service	Using Odata for Create, Read, Delete, and Update (CRUD) function for the logistic system
Debugging	Faced issues with database table creation, especially draft-related tables. These were resolved by correcting the database configuration in package.json to point to the correct db.sqlite file.

Table 3.3.1 Step of trying the sap btp

3.4 What you learned from the process

I had learned SAP CAP , SAP Fiori and Odata service and core data service which divide to database layer and service layer. I learn SAP by viewing YouTube videos, SAP documentation, and the official SAP tutorial, then apply it in SAP. It is an interesting journey. I also found that database tables aren't created for you and you have to do it manually. Moreover, I had also learned how to draft some mechanism in SAP manages the temporary data before it gets saved.